

Ansible mein automation ke do tareeqay

Ansible automation ko ek martaba chalne wali commands ya dobara istemal honay wali YAML files ke zariye chala sakta hai.

1 Command ka tareeqa

Ansible ad-hoc commands

Tez aur seedhay tasks ke liye behtar hai jahan reusable workflow ki zaroorat na ho.

2 File ka tareeqa

Ansible playbooks

Munazzam aur bar bar chalne wali automation ke liye behtar hai jo scale ho aur asani se maintain ki ja sake.

Ad-hoc commands ki pabandiyan

- Jab bhi task dobara karna ho, command phir se type karni parti hai.
- Bari automation jobs ke liye kai alag commands chalani parti hain.
- Mushkil conditions ko likhna aur maintain karna asan nahin hota.
- Bar bar input dene wale kaam, jaise kai users banana, alag commands mangte hain.
- Playbooks ke muqablay mein variables aur reusable structure limited hota hai.

Ansible playbooks ke faiday

- Har command dobara type karne ke bajaye tasks ko file mein save karein.
- Kai automation tasks ko ek munazzam playbook mein jama karein.
- Jab action system ki state par depend kare to conditional logic lagayen.
- Ek workflow mein kai inputs aur tasks ko manage karein.
- Automation ko flexible aur reusable banane ke liye variables istemal karein.

Ansible playbook ki bunyadi baatein

- Playbook mein woh tasks hotay hain jo Ansible managed nodes par perform karta hai.
- Ansible playbooks YAML syntax istemal karte hain.
- Playbook ko kisi bhi aisi jagah rakha ja sakta hai jahan user usay access kar sake.
- Filename koi bhi ho sakta hai, lekin extension .yaml ya .yml honi chahiye.
- YAML playbooks ko parhne aur review karne mein asan banata hai.

playbook.yml

```
---  
- name: Install and start  
  Apache  
  hosts: webservers  
  become: true  
  tasks:  
    - name: Install httpd  
      dnf:  
        name: httpd  
        state: present
```

YAML aur Ansible playbooks

YAML configuration aur automation ko insani tor par parhne layak andaaz mein bayan karta hai.

- Kai references YAML ko Yet Another Markup Language kehte hain.
- Is ka official recursive naam YAML Ain't Markup Language hai.
- YAML structured data ko represent karta hai jo configuration tools istemal karte hain.
- Ansible playbooks aur mutaliqa data files ke liye YAML istemal karta hai.
- AWS CloudFormation aur Kubernetes bhi YAML documents istemal karte hain.

YAML likhne ke bunyadi qawaid

- YAML document --- se shuru aur ... par khatam ho sakta hai.
- List ka har item hyphen aur us ke baad space se shuru hota hai.
- Mutaliqa items ko barabar indentation dein. YAML indentation ke liye tabs istemal nahin karta.
- Key-value data mein colon ke baad space hoti hai: key: value

Data ki aam forms

String • List • Mapping

YAML structure

```
---  
- name: Install packages  
  hosts: webserver  
  tasks:  
    - name: Install httpd  
      dnf:  
        name: httpd  
        state: present  
...
```

YAML string format

String format mein ek key ke neeche ek value store hoti hai.

Single key-value pairs

```
Name: Khalid  
hosts: webserver  
user: root  
package: httpd
```

Zaati misaal

Naam	Khalid
------	--------

Format	key: value
--------	------------

YAML array aur list format

List ek hi key ke neeche kai values store karti hai. Har item hyphen aur space se shuru hota hai.

Packages ki list

```
package:  
  - httpd  
  - vsftpd  
  - firewallld
```

Indentation note

Tamam list items ko ek hi indentation level par rakhein. Barabar style YAML ko parhna asan banata aur syntax errors se bachata hai.

YAML mapping format

Mapping mutaliqa subkeys aur values ko ek parent key ke neeche group karti hai.

Zaati profile mapping

```
Khalid:
  Phone: +17732222333
  Address: Chicago
  Skills:
    - Linux
    - Java
    - Flutter
```

Structure

Parent key	Khalid
Subkeys	Phone, Address, Skills
Qaidah	Child data ko barabar indent karein

YAML se Ansible playbook tak

YAML structure deta hai. Ansible keywords aur modules automation ko define karte hain.

YAML

Indentation
Lists
Mappings

Play

Hosts
Privileges
Task list

Tasks

Modules
Arguments
Desired state

Play hosts ko target karta hai. Har task arguments ke sath ek module call karta hai.

Playbook ka structure

nginx.yml

```
---
- name: Configure web servers
  hosts: web
  become: true

  tasks:
    - name: Install Nginx
      dnf:
        name: nginx
        state: present
```

Play

Automation ka ek unit jo host pattern ko target karta hai.

Task

Tasks list ke andar ek naam diya gaya action.

Module

Ansible ka tool jo action perform karta hai, jaise dnf.

Arguments

Module ko di janay wali values, jaise name aur state.

Pehla Nginx playbook

nginx.yml

```
---
- name: Install and start Nginx
  hosts: web
  become: true

  tasks:
    - name: Install Nginx
      dnf:
        name: nginx
        state: present

    - name: Start and enable Nginx
      service:
        name: nginx
        state: started
        enabled: true
```

Nateeja

- Web inventory group ko target karta hai
- become ke zariye sudo istemal karta hai
- nginx package install karta hai
- nginx start karta aur boot par enable karta hai

Play-level keywords

```
---
```

YAML document ka optional marker

```
- name
```

Play ki asani se parhi janay wali wazahat

```
hosts: web
```

Play ko web inventory group par chalata hai

```
become: true
```

Administrative tasks ke liye privilege escalation istemal karta hai

```
tasks:
```

Tasks ki tarteeb wali list shuru karta hai

Task, module aur arguments

Ek task

```
- name: Install Nginx
  dnf:
    name: nginx
    state: present
```

- name batata hai ke task ko kya hasil karna hai.
- dnf package-management module select karta hai.
- name: nginx package ki pehchan karta hai.
- state: present yaqini banata hai ke package installed ho.

Matlooba state

Ansible current system ko matlooba state se compare karta aur sirf zaroori tabdeeli karta hai.

Playbook ki jaanch

Managed nodes par tabdeeli se pehle checks chalayen.

1. Syntax check

```
ansible-playbook --syntax-check  
nginx.yml
```

YAML aur playbook syntax check karta hai.
Normal task execution ke liye managed nodes se
rabta nahin karta.

2. Check mode

```
ansible-playbook nginx.yml --check
```

Kai tabdeeliyon ka andaza lagata hai magar
apply nahin karta. Kuch modules har operation ko
mukammal simulate nahin karte.

Optional host check

```
ansible-playbook nginx.yml --list-hosts
```

Playbook chalana

hosts: web se match honay wale tamam hosts par chalayein

```
ansible-playbook nginx.yml
```

Selected host pattern mein sirf node1 par chalayein

```
ansible-playbook nginx.yml --limit node1
```

--limit ehtiyat se istemal karein. Limited host pehle se play ke host pattern ka hissa hona chahiye.

Idempotency

Achha likha hua playbook bar bar chal sakta hai aur ghair zaroori tabdeeliyan nahin karta.

Pehli run

changed=2

Nginx install aur start hota hai.

Doosri run

changed=0

Matlooba state pehle se mojood hai.

Playbook ke baad tasdeeq

Service check karein

```
ansible web -m command \  
-a "systemctl is-active nginx"
```

Mutawaqqa service result

active

Web response check karein

```
ansible web -m uri \  
-a "url=http://localhost  
status_code=200"
```

Mutawaqqa HTTP result

status: 200

Verification service ki state confirm karti aur check karti hai ke Nginx HTTP request ka jawab deta hai.

Playbook ki aam ghaltiyan

- Tabs ya ghair barabar indentation istemal karna
- Colon ke baad space bhool jana
- Aisay inventory group ko target karna jo mojood nahin
- Module arguments ko ghalat indentation level par rakhna
- Package aur service administration ke liye become: true bhool jana
- Aisa module istemal karna jo managed node ke operating system se match na kare

Tajweez karda workflow

1. Syntax check karein
2. Target hosts review karein
3. Check mode istemal karein
4. Playbook chalayen
5. Nateeja verify karein